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Tunable nonlinear optical absorption in amorphous and crystalline Sb_2Se_3 thin films

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ABSTRACT

The nonlinear optical absorption properties of as-deposited and crystalline Sb_2Se_3 films were systematically studied by using open-aperture Z-scan technique with femtosecond laser pulses at the wavelength of 800 nm. It is interesting to find that the crystallization leads nonlinear optical absorption to switch from reverse saturable absorption to saturable absorption. A tunable behavior is attributed to the competition between ground state excitation and excited state absorption, which depends critically on the facts of band gap energy and the density of localized defect states in the Sb_2Se_3 film. A three-level approximate model is proposed to explain the probable electronic transition and its rationality is further confirmed by second harmonic experiments under the same conditions. In particular, it is found that there are two different nonlinear optical absorption mechanisms coexisting in one kind of material with identical chemical composition, and their switching behavior is regulated merely by phase-change after thermal treatment. These special properties of Sb_2Se_3 thin film imply huge potential applications in the field of innovative nonlinear optical devices.

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1. Introduction

Nonlinear optical (NLO) materials have attracted much research interest in past decades owing to their strong SHG/THG response, large laser damage threshold, larger nonlinear absorption coefficient and easy fabrication of high quality samples, which are expected as promising materials for various civilian and military applications, such as deep-ultraviolet light generation, mode-locking, super-resolution imaging and optical limiting [1–8]. Especially, tremendous progress has been made in the investigations of their nonlinear optical absorption (NOA) properties since the discovery of Z-scan technique. According to the number of photons absorbed by one absorption process in the medium, the NOA can be simply divided into two types of single-photon absorption and multi-photon absorption. Single-photon NOA can be further classified into saturable absorption (SA) and reverse saturation absorption (RSA). Both SA and RSA are results of the dynamical interaction of the material with the incident laser. For SA, the absorption cross section of the excited state is smaller than that of the ground state. The electrons in SA material can transfer from the ground state to an upper energy state by absorbing incident photons, thus, the ground state population will be depleted. In this case, the transmittance increases as the incident laser intensity increases. Conversely, for RSA, the absorption cross section of the excited state is larger than that of the ground state. Then the transmittance decreases with increasing incident laser intensity owing to the excited state transitions [9,10].

Although several methods are used to improve the NOA characteristics in recent years, many traditional NOA materials are based on SA or RSA independently of each other, which limits their actual applications to a great extent [11–15]. Hence, considerable efforts have been devoted to design new-type nonlinear optical devices using SA and RSA simultaneously. Philip et al. proposed a passive all-optical diode with semiconductor doped glass-copper phthalocyanine (CuPc) solution where the glass was one kind of strong SA material, but CuPc was RSA one [10]. For this device, the main drawbacks that expected to be avoided in large-scale

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integration (LSI) were the thickness of glass and the use of solvent. Anand et al. presented the first solid-state all-carbon passive optical diode by employing graphene (SA) and C_{60} (RSA) in tandem [9]. This innovative design overcame the above drawbacks and it was more suitable for LSI. In spite of all-optical diode, Roy et al. theoretically analyzed ultrafast SA to RSA transitions in graphene oxide thin films and designed all-optical parallel logic gates based on the conversion between the two mechanisms [16]. Most of the mentioned designs require combining different materials that exhibit SA or RSA, respectively. In this respect, it will be advantageous in the realization of SA to RSA transition in one kind of material with identical chemical composition. Coincidentally, our research group has been working on finding such materials. As reported in our previous work, J. Wang et al. found that the transition of SA to RSA could be observed in In_2Te_3 thin film only by adjusting its thickness [17]. Therefore, we predict that similar chalcogenide has interesting NOA properties.

As a member of chalcogenides, binary thin films with layered structure in the groups V-VI have simply composition with fixed orthorhombic phase [18,19]. Among them, antimony selenide (Sb_2Se_3) is a promising direct band gap absorber with a high absorption coefficient. On the basis of the excellent optical and electrical properties [20–22], Sb_2Se_3 thin films are expected to be suitable for PCRAM [23], photodetectors [24], photovoltaics [25] and so on. However, the above-mentioned fundamental principles are all linear optical and electrical properties, there are only a few researches study the nonlinearity. Interestingly enough, Sb_2Se_3 in different forms present different nonlinear properties. Molli et al. reported that Sb_2Se_3 nanoparticles exhibit strong intensity-dependent NOA response which was found to be two-photo absorption (2 PA) or three-photo absorption (3 PA) process [26]. On the contrary, Yadav et al. found that one-dimensional Sb_2Se_3 nanowires showed SA response in the excitation of both ns and fs pluses [27]. To the best of our knowledge, there have been no studies on the nonlinearity of Sb_2Se_3 thin films in literature. Furthermore, it is proved that the band gap of amorphous Sb_2Se_3 (a-Sb_2Se_3) thin film is larger than that of crystalline Sb_2Se_3 (c-Sb_2Se_3) thin film and the tunable behavior from RSA to SA can be adjusted by the band gap of the material [28]. We infer that crystallization may affect the nonlinear properties of Sb_2Se_3 thin film. In this study, interesting tunable behavior of NOA type in Sb_2Se_3 thin films was achieved only by thermal treatment. This interesting phenomenon provides a simple way to design innovative nonlinear optical devices.

2. Experimental

The a-Sb_2Se_3 thin films were deposited on fused quartz substrates by radio-frequency (RF) magnetron sputtering where the sputtering target was Sb_2Se_3 with purity of 99.99%. The background and working pressures were 7×10^{-6} mbar and 2.8×10^{-3} mbar in the chamber, respectively. The deposition power of the Sb_2Se_3 target was fixed at 60 W and the deposition time was set up to 200 s. Meanwhile, the substrates were maintained at room temperature during sputtering process. To obtain c-Sb_2Se_3, the a-Sb_2Se_3 film was annealed at 300 °C for 30 min and then cooled down to room temperature in flowing nitrogen ambient.

The spectroscopic ellipsometry (SE) was used to determine the thicknesses and linear optical properties of the samples via Lorentz oscillator model in the wavelength range of 380–800 nm. All optical constants of the samples were obtained by fitting raw data using a commercial software FilmWizard®. Structural characteristics of the samples were determined by X-ray diffraction (XRD) with Cu-Kα (1.54056 Å) radiation (Bruker D8 ADVANCE) in a 2θ range of 10–60° with a step 0.02°. A double-beam UV-VIS-NIR spectrophotometer (Shimadzu UV-3600) was applied to characterize the transmission and absorption spectroscopic characteristics.

To measure the NOA properties of Sb_2Se_3 films, we employed a single beam open-aperture (OA) mode Z-scan setup which was discussed in Ref. [17]. In the experiments, 800 nm (fundamental) and 400 nm (second harmonic) wavelengths with 100 fs pulse duration and 1 kHz repetition rate from the Ti-sapphire regenerative amplifier system (Spectra Physics, Spitfire Ace) were applied to excite the samples. The beam radius ω at the focal point and the Rayleigh length ($Z_0 = \frac{\pi\omega^2}{\lambda}$) were 32 μm and 4 mm, respectively. The samples were moved along the propagation direction (z axis) using a stepping motor. Each thickness of the samples was much smaller than the diffraction length of focused beam, so the samples could be regarded as 'thin' films [29]. After testing, the system met the requirements of Z-scan theory [17].

3. Results and discussion

The thicknesses of the a-Sb_2Se_3 and c-Sb_2Se_3 thin films determined from ellipsometric data fitting are 52.2 nm and 51.2 nm, respectively. Fig. 1 shows the XRD patterns of the as-deposited and annealed samples. For Sb_2Se_3 thin film deposited at room temperature, no diffraction peaks are observed due to the amorphous nature. After annealing at 300 °C, strong diffraction peaks appear obviously, and all the diffraction peak positions agree well with the standard diffraction pattern of orthorhombic phase c-Sb_2Se_3 (JCPDS No: 015-0861) with lattice parameters a, b, and c are 11.63, 11.78 and 3.98 Å, respectively. Because the crystalline temperature from amorphous phase to orthorhombic phase is about 133 °C [30], it can be confirmed that the thermal treatment of 300 °C has made the a-Sb_2Se_3 thin film transform to c-Sb_2Se_3 thin film completely based on the experimental result.

The optical constants of a-Sb_2Se_3 and c-Sb_2Se_3 thin films were measured by SE, a precise nondestructive apparatus to fit optical parameters [31]. The fitting results are portrayed in Fig. 2(a) and (b). It is clear that both refractive index n and extinction coefficient k of a-Sb_2Se_3 and c-Sb_2Se_3 thin films are very large in the visible spectrum, indicating that Sb_2Se_3 thin film has large imaginary part of the third-order nonlinear optical susceptibility and strong linear absorption [32]. Moreover, the optical constant curves of c-Sb_2Se_3 thin film are similar to those of a-Sb_2Se_3 thin film, while the former sample has higher values. It is mainly because the unsaturated defects gradually decrease and grain sizes increase during annealing process [33]. The acquisition of optical constants not only gives

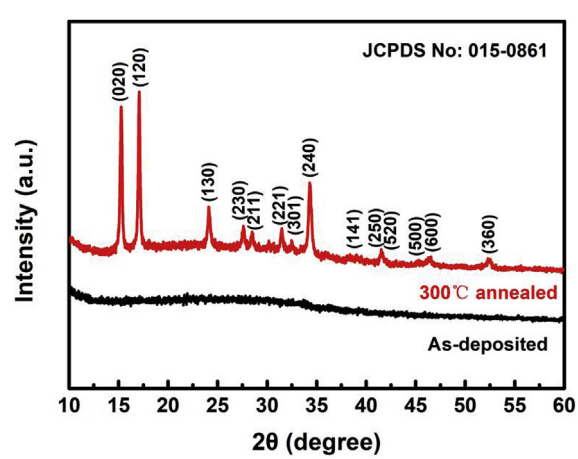

Fig. 1. XRD patterns of a-Sb_2Se_3 and c-Sb_2Se_3 thin films.

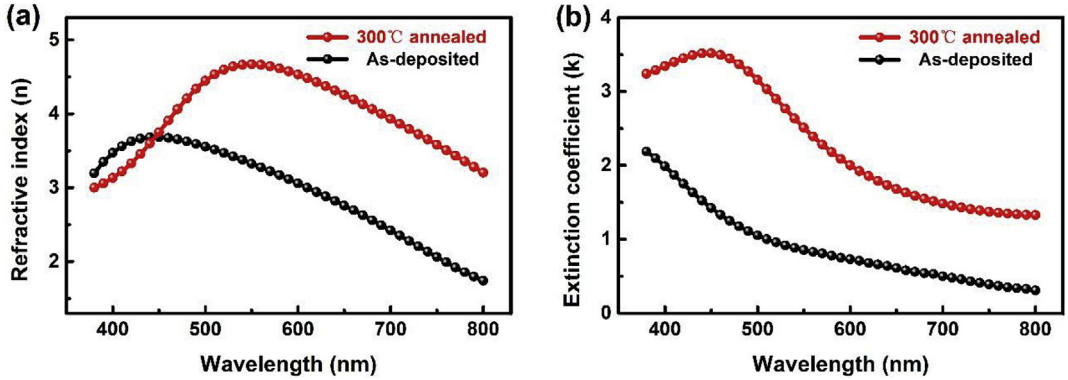

Fig. 2. (a) Refractive index n and (b) extinction coefficient k curves of Sb_2Se_3 thin films.

us a better understanding of the properties of Sb_2Se_3 material but also provides parameters for the following NOA investigation.

Fig. 3 displays the optical transmission spectra and linear optical absorption spectra of the a-Sb_2Se_3 and c-Sb_2Se_3 thin films in the spectrum range of 300–1600 nm. In Fig. 3(a), it is found that the transmittance of a-Sb_2Se_3 thin film is 80.5%, much larger than that (12.2%) of c-Sb_2Se_3 thin film at the wavelength of 800 nm. While the transmittance of both a-Sb_2Se_3 and c-Sb_2Se_3 thin films are almost zero at 400 nm wavelength. Moreover, the samples do not show any interference pattern whereas the absorption edge of c-Sb_2Se_3 thin film shift towards longer wavelength (red-shift), which indicates that the band gap becomes narrow after thermal treatment.

To determine optical band gap and NOA mechanism, the linear absorption coefficient α is a key parameter. Since Sb_2Se_3 has a high linear absorption coefficient ($>10^5$ cm^{-1}) [34], the condition $\alpha d \approx 1$ is fulfilled. Thus, the α can be calculated from the equation $\alpha = -ln(T)/d$, where T is the transmittance, d is the thickness of Sb_2Se_3 thin film [22,30]. The calculated linear absorption spectra are plotted in Fig. 3(b). On the one hand, Sb_2Se_3 thin films have strong linear absorption and the linear absorption coefficient of c-Sb_2Se_3 thin film is larger than that of a-Sb_2Se_3 thin film, indicating that thermal treatment can improve linear optical absorption. Such strong absorption characteristic of Sb_2Se_3 makes it a promising candidate for excellent solar energy material [35]. On the other hand, the spectra characteristics indicate that the absorption region covers the entire visible region and extends up to near infrared region. This observed feature can be attributed to the presence of different density of localized defect states in the forbidden gap. Optical absorption behaviors largely depend on these defect states since they can affect the transitions of electrons.

Fig. 4 exhibits how to estimate the optical band gaps for the a-Sb_2Se_3 and c-Sb_2Se_3 thin films by the linear extrapolation of the absorption coefficients. Since Sb_2Se_3 film is a direct band gap semiconductor material, according to the Tauc's method [36], the optical band gaps (E_g) of the thin films can be calculated by using the following equation,

$$(\alpha h\nu)^2 = A(h\nu - E_g) \quad (1)$$

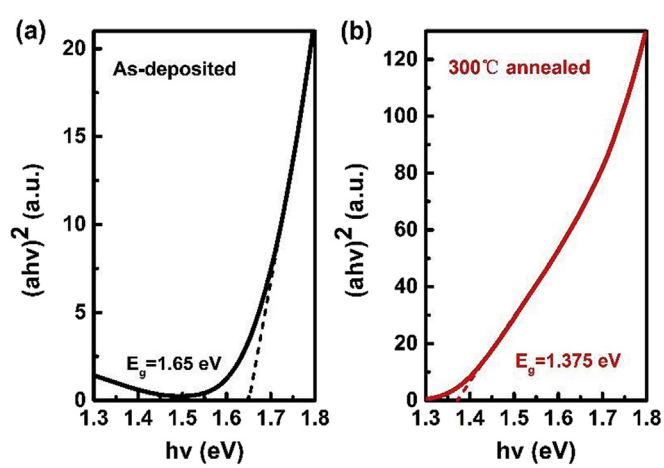

Fig. 4. The optical band gaps of (a) a-Sb_2Se_3 and (b) c-Sb_2Se_3 thin films.

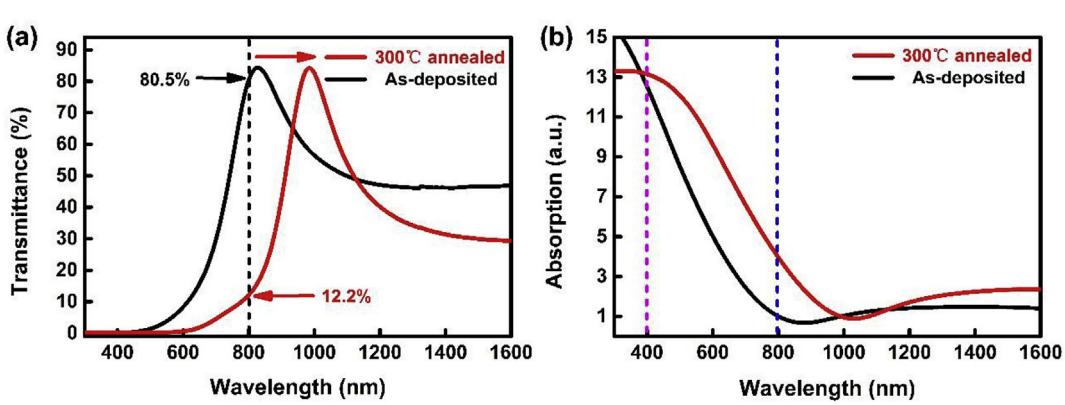

Fig. 3. The (a) transmission and (b) absorption spectra of a-Sb_2Se_3 and c-Sb_2Se_3 thin films.

where α, $h\nu$, and A are the linear absorption coefficient, photon energy, and a constant which related to effective mass, respectively. The optical band gap (E_g) can be derived from the interception of the linear part of $(\alpha h\nu)^2$ *vs.* $h\nu$ for $(\alpha h\nu)^2 = 0$. Based on formula (1), the optical band gaps are calculated as 1.650 eV for a-Sb_2Se_3 thin film and 1.375 eV for c-Sb_2Se_3 thin film. Obviously, the optical band gap of Sb_2Se_3 thin film decreases after annealing, corresponding to the red-shift of transmission spectra.

Single beam OA Z-scan was employed to study the NOA properties of Sb_2Se_3 thin films with femtosecond laser pulses at the wavelength of 800 nm [17]. The OA Z-scan system measured the intensity-dependent transmittance while the films were sequentially moved along the Z-axis. In order to avoid nonlinear scattering, the samples were performed at two different relative low incident power densities of 66 GW/cm^2 and 99 GW/cm^2. It is worth noting that the transmittances of the fused quartz substrates are almost unchanged during the experiment, thus the NOA behaviors only originate from Sb_2Se_3 thin films. The OA Z-scan results are presented in Fig. 5. Clearly, the NOA behavior of the Sb_2Se_3 thin film is highly sensitive to the thermal treatment. As shown in Fig. 5(a), the normalized transmittance decreases when the position of a-Sb_2Se_3 thin film approaches the focal point, leading to a symmetric valley in the Z-scan curve. Herein, the valley shape belongs to RSA response. In contrast to a-Sb_2Se_3, the c-Sb_2Se_3 thin film exhibits a symmetric peak, i.e., the normalized transmittance shows a gradual increase which indicating SA response given in Fig. 5(d). Moreover, both a-Sb_2Se_3 and c-Sb_2Se_3 thin film have larger modulation depth (valley depth or peak height) under the excitation of higher incident power density (99 GW/cm^2).

To depict device potential of the Sb_2Se_3 thin films, the output intensities (I_{out}) are plotted as a function of the input intensities (I_{in}) in Fig. 5(b) and (e). Obviously, the slope of $I_{out} - I_{in}$ curve of a-Sb_2Se_3 thin film gradually decreases, indicating an optical limiting effect. By contrary, the slope of $I_{out} - I_{in}$ curve of the c-Sb_2Se_3 thin film is gradually increasing, which suggests that the transmittance is increased. Concisely, whether a-Sb_2Se_3 or c-Sb_2Se_3 thin films, I_{out} is no longer linear with I_{in} when the latter increases to a higher incident power density. In addition, the a-Sb_2Se_3 and c-Sb_2Se_3 thin

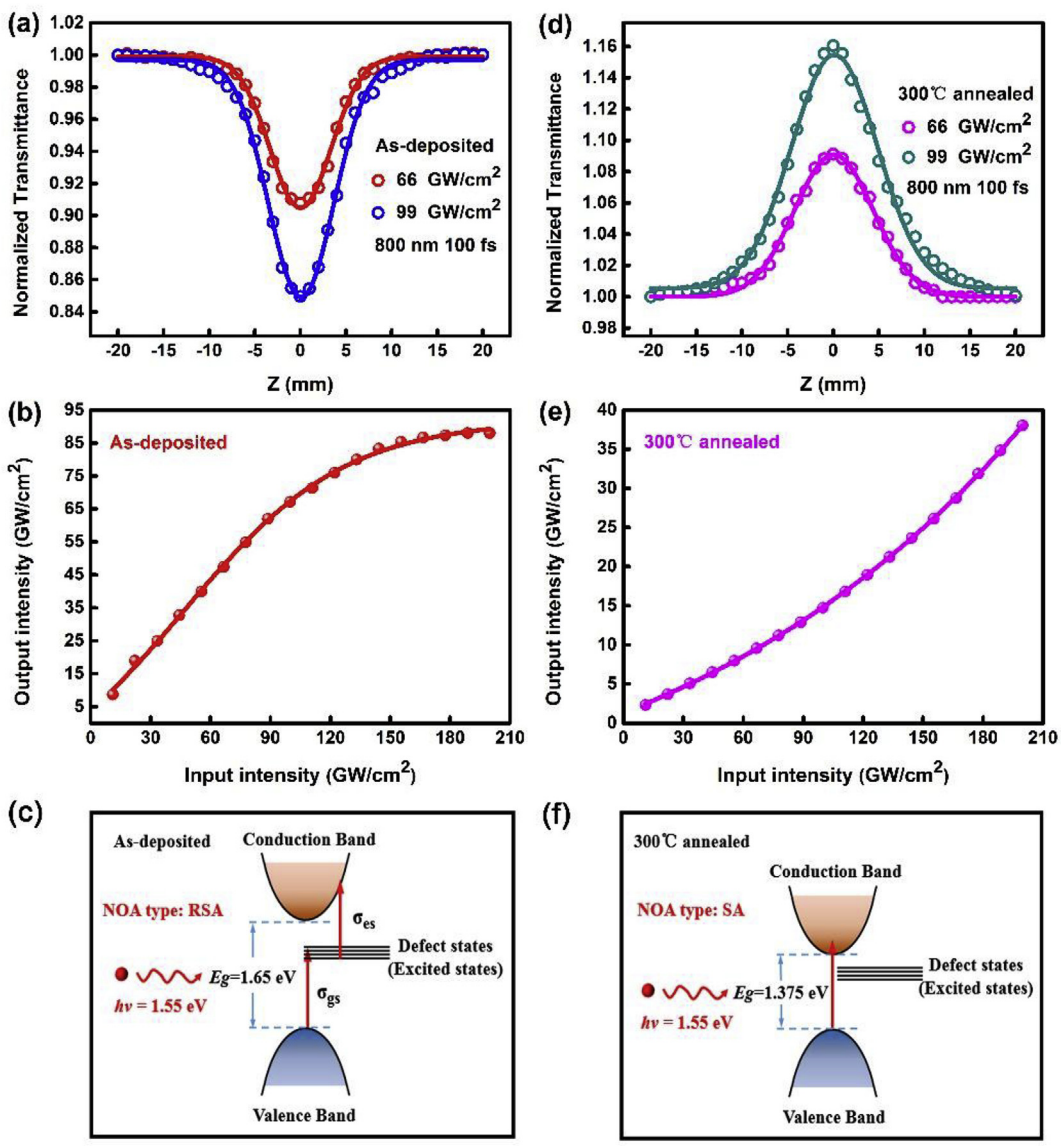

Fig. 5. At the excitation of 800 nm wavelength, (a) OA Z-scan curves of a-Sb_2Se_3 thin film. (b) $I_{out} - I_{in}$ curve of a-Sb_2Se_3 thin film. (c) The three-level model of a-Sb_2Se_3 thin film. (d) OA Z-scan curves of c-Sb_2Se_3 thin film (e) $I_{out} - I_{in}$ curve of c-Sb_2Se_3 thin film and (f) the three-level model of c-Sb_2Se_3 thin film.

films present nearly linear behavior at low I_{in}, but exhibit RSA and SA at the higher intensity, respectively. The distinctive tunability of NOA behavior in Sb_2Se_3 thin film can be applied to design all-optical devices which combining RSA with SA materials simultaneously [9].

Based on the experimental results, it is observed that the RSA-to-SA switching has been realized by thermal treatment, which is different from the single NOA type of Sb_2Se_3 material surveyed in previous reports [26,27]. We propose a three-level model to explain the probable electronic transition of the tunable behavior. It is known that unsaturated bonds of Sb_2Se_3 samples are responsible for the formation of localized defect states in the thin films. In particular, the a-Sb_2Se_3 thin film has more localized defect states comparing to the c-Sb_2Se_3 thin film. These defect states assist electrons transiting from valance band to conduction band. As portrayed in Fig. 5(c), the optical band gap of a-Sb_2Se_3 thin film (1.650 eV) is larger than the excited photon energy (~1.55 eV). In this condition, electrons can transfer from valence band to defect states by absorbing excited photons. At the same time, some of the electrons in the defect states can be transferred to higher excited states in the conduction band by absorbing other photons. Similar NOA behavior also appeared in the In_2Te_3 thin film with thicker thickness [17]. It is well known that thermal treatment is a common method to reinforce crystallization and release unsaturated defects. Therefore, c-Sb_2Se_3 thin film has a lower optical band gap (1.375 eV) comparing to a-Sb_2Se_3 thin film. In this point, the optical band gap (1.375 eV) of c-Sb_2Se_3 thin film is less than the excited photon energy (~1.55 eV). Fig. 5(f) presents the three-level model of c-Sb_2Se_3 thin film. The electrons can directly transfer from valence band to conduction band by absorbing excited photons, then the conduction band becomes almost occupied and the Pauli-blocking avoids further absorption of photons, thereby resulting in the SA response.

For the purpose of demonstrating NOA type more intuitively, the ground state (σ_{gs}) and the excited state (σ_{es}) absorption cross sections are calculated by the following two equations [27,37].

$$\sigma_{gs} = -\frac{logT_0}{NL} \tag{2}$$

$$\sigma_{es} = -\frac{logT_{max}}{NL} \tag{3}$$

where N is the absorber density, L is the sample thickness, T_0 is the transmission in the linear region and T_{max} is the maximal transmission at high intensity. RSA occurs when the σ_{es} is much larger than the σ_{gs}, in turn, SA will occur if the σ_{gs} is much larger than the σ_{es}. The calculated results are listed in Table 1.

For a-Sb_2Se_3 thin film, both the ratio of $\sigma_{es}/\sigma_{gs} = 2.21$ at 66 GW/cm^2 and that of 2.48 at 99 GW/cm^2 are much larger than 1, indicating that a-Sb_2Se_3 thin film is a RSA material. Whereas for c-Sb_2Se_3 thin film, the ratio of $\sigma_{es}/\sigma_{gs} = 0.80$ at 66 GW/cm^2 and that of 0.78 at 99 GW/cm^2 are much smaller than unity, resulting in the SA phenomena. These calculation results offer the evidences of experimental phenomena, that is, RSA response of a-Sb_2Se_3 thin film and SA response of c-Sb_2Se_3 thin film, respectively.

In terms of fitting calculation, the typical OA Z-scan theory which proposed by Sheik-Bahae is used to fit the raw data. Our calculations mainly focus on the NOA coefficient of a-Sb_2Se_3 and c-Sb_2Se_3 thin films at the wavelength of 800 nm. The nonlinear effective absorption coefficient β_{eff}, which defined as $\alpha = \alpha_0 + \beta_{eff} I$ can be deduced by numerically fitting the following equation [29],

$$T_{OA} = \sum_{m=0}^{\infty} \frac{\left[-\beta_{eff} I_0 L_{eff} \Big/ \left(1 + z^2/z_0^2\right)\right]^m}{(m+1)^{3/2}} \tag{4}$$

where T_{OA} is the normalized transmittance of the OA Z-scan measurement, I_0 is the incident laser intensity at the focal point, z is the position of the samples on the axis, z_0 is the Rayleigh length and L_{eff} is the effective thickness of samples which can be calculated by Eq. (5) [38],

$$L_{eff} = \frac{1 - e^{-\alpha_0 L}}{\alpha_0} \tag{5}$$

where α_0 is the linear absorption coefficient at 800 nm wavelength, L is the physical thickness of the sample. The imaginary part of the third-order nonlinear optical susceptibility ($\mathrm{Im}\chi^{(3)}$) is directly related to β_{eff} by the following Eq. (6) [39],

$$\mathrm{Im}\chi^{(3)} = \left[\frac{10^{-7} c\lambda n^2}{96\pi^2}\right]\beta_{eff} \tag{6}$$

where $\mathrm{Im}\chi^{(3)}$ is in esu, β_{eff} is in cm/W; c, λ and n are the light speed in vacuum, wavelength of the incident laser and the linear refractive index of the sample, respectively. A figure of merit for the third order optical nonlinearity is defined as FOM $= \left|\mathrm{Im}\chi^{(3)}/\alpha_0\right|$. It is used to eliminate the discrepancy caused by the linear absorption α_0. Moreover, the α_0 of each sample is different, so we mainly take FOM as a norm to evaluate the SA performance of the thin film [32]. To obtain the saturation intensity I_s, one can use the propagation equation which is given by Ref. [29],

$$\frac{dI}{dz} = -\frac{\alpha_0 I}{1 + I/I_s} \tag{7}$$

here I, z, and α_0 are the incident intensity, propagation distance, and the linear absorption coefficient. All NLO parameters are summarized in Table 2.

In Z-scan experiment, the material exhibits a valley indicating RSA if $\beta_{eff} > 0$ or it exhibits a peak indicative of SA if $\beta_{eff} < 0$. After annealing, it is clear that the β_{eff} of Sb_2Se_3 thin film has increased significantly and the sign of the β_{eff} also switches from positive to negative, representing that NOA mechanism switches from RSA to SA. The distinctive property, nonlinear absorption coefficient tuned by annealing, is of great application in the optical field, such as electro-optic routers [40], mid-IR all optical switching devices [41] and so on. At 800 nm fs pulse excitation, the β_{eff} of the a-Sb_2Se_3 and c-Sb_2Se_3 thin films are much larger than those of In_2Te_3 and InTe thin films which have been investigated by our team [4,17]. The FOMs of all the sample films have the same magnitude of

Table 1
NOA types of the a-Sb_2Se_3 and c- Sb_2Se_3 thin films measured by OA Z-scan.

Sample	Thickness (nm)	Intensity (GW/cm^2)	σ_{es}(cm^2)	σ_{gs}(cm^2)	σ_{es}/σ_{gs}	NOA type
As-deposited	52.2	66	4.09×10^{-18}	1.93×10^{-18}	2.21 > 1	RSA
		99	4.88×10^{-18}	1.97×10^{-18}	2.48 > 1	RSA
300 °C annealed	51.2	66	1.95×10^{-17}	2.44×10^{-17}	0.80 < 1	SA
		99	2.00×10^{-17}	2.55×10^{-17}	0.78 < 1	SA

Table 2
NLO parameters of the a-Sb_2Se_3 and c-Sb_2Se_3 thin films measured by OA Z-scan.

Sample	β_{eff} (cm/GW)	Im$\chi^{(3)}$ (esu)	FOM (esu·cm)	I_s (GW/cm^2)
As-deposited	843.67	6.47×10^{-17}	1.55×10^{-21}	N/A
300 °C annealed	−1831.01	-4.76×10^{-16}	1.16×10^{-21}	158.82

10^{-21} esu·cm, but fall slightly after annealing. It is noticed that the calculated value of saturation intensity I_s for c-Sb_2Se_3 thin film is 158.82 GW/cm^2, which is consistent with the former study of Sb_2Se_3 nanowires [27]. In addition, it is found that the Sb_2Se_3 thin films have better NLO property, namely larger β_{eff}, FOM and lower I_s. Based on these characteristics, a-Sb_2Se_3 and c-Sb_2Se_3 thin films have the potential for applications of optical power limiter and passive mode-locker [32].

Generally, the RSA-to-SA switching behavior is considered as the consequence of the competition between the ground state excitation and the excited state absorption. It mainly depends on the excited photon energy and the band gap energy of material [42]. Importantly, the density of localized defect states plays an important role in the process of electronic transition. In order to confirm the three-level model, additional OA Z-scan experiments by using excitation wavelength of 400 nm were performed at the same incident power density. It can be seen in Fig. 6(a) and (d), unlike the tunable behavior, the normalized transmittances of a-Sb_2Se_3 and c-Sb_2Se_3 thin films increase as the samples move toward the beam focus, showing clear SA responses. The SA responses become much more pronounced as the incident energy increases. Either the slope of $I_{out} - I_{in}$ curve of a-Sb_2Se_3 thin film or that of c-Sb_2Se_3 thin film is gradually increasing, as shown in Fig. 6(b) and (e). The results reveal that the transmittance is highly dependent on

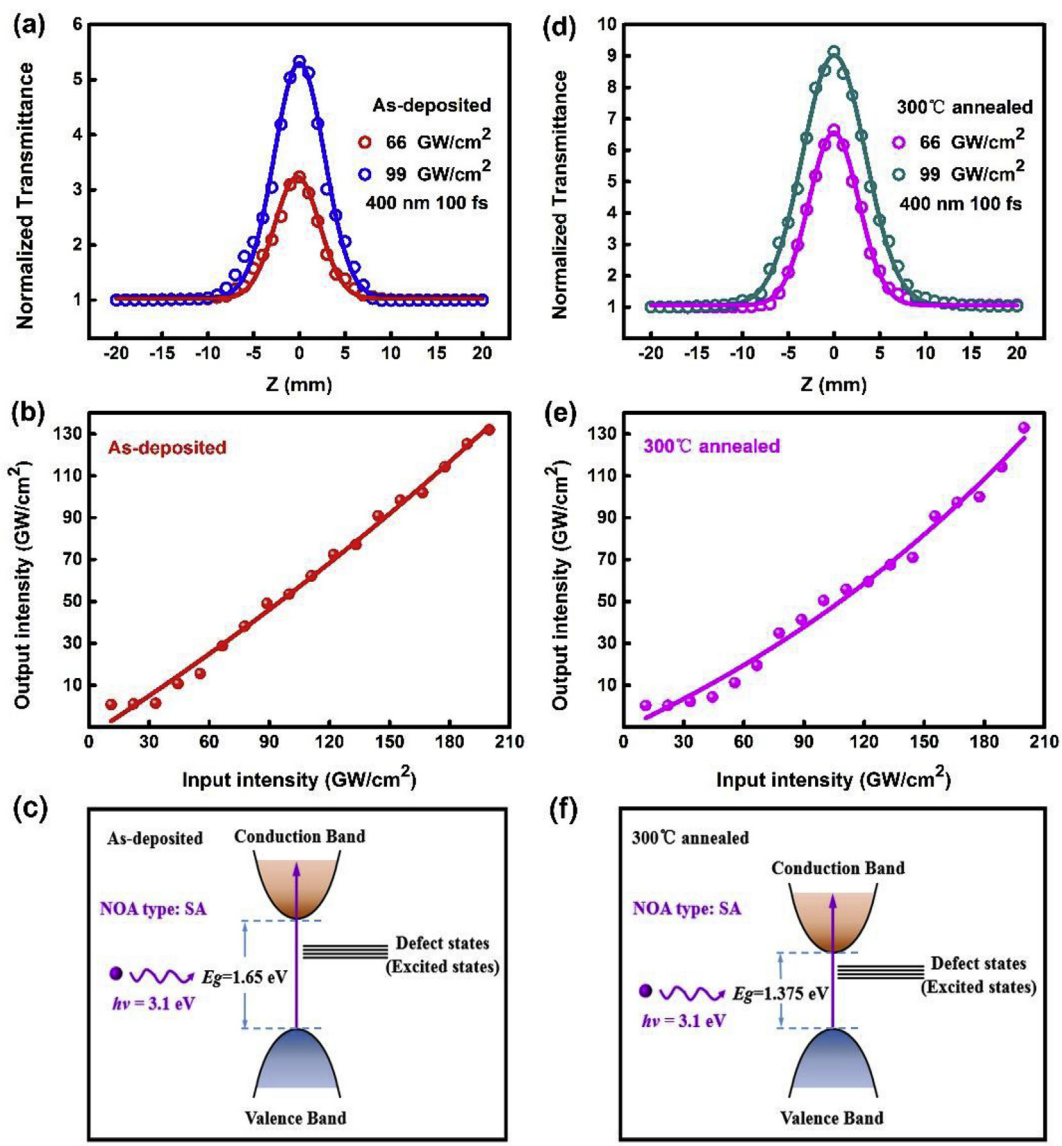

Fig. 6. At the excitation of 400 nm wavelength, (a) OA Z-scan curves of a-Sb_2Se_3 thin film. (b) $I_{out} - I_{in}$ curve of a-Sb_2Se_3 thin film. (c) The three-level model of a-Sb_2Se_3 thin film. (d) OA Z-scan curves of c-Sb_2Se_3 thin film (e) $I_{out} - I_{in}$ curve of c-Sb_2Se_3 thin film and (f) the three-level model of c-Sb_2Se_3 thin film.

the incident power density. The stronger the incident laser power density, the stronger the transmittance, indicating a typical SA behavior. Similarly, Fig. 6(c) and (f) depict the three-level models of a-Sb_2Se_3 and c-Sb_2Se_3 thin films, respectively. Because the energy of excited photons (~3.10 eV) are much larger than the optical band gaps of studied materials (1.650 eV for a-Sb_2Se_3 and 1.375 eV for c-Sb_2Se_3), under these intensities, the ground state carries are depleted. As a result, the optical transition is reduced and SA behavior appears.

4. Conclusions

In summary, the NOA responses of a-Sb_2Se_3 and c-Sb_2Se_3 thin films have been investigated by using OA Z-scan technique in the femtosecond excitation regime. The results demonstrate a thermal effect-related NOA tunability in Sb_2Se_3 thin films. That is to say, NOA behavior at the excitation wavelength of 800 nm can be switched from RSA to SA by thermal treatment, corresponding to the amorphous (as-deposited) to crystalline (after annealing) phases of the sample films. The interesting switching behavior is the consequence of the competition between the ground state absorption and the excitation state absorption, which depends critically on the facts of band gap energy and the density of localized defect states in the film. Further, all the films show SA responses at the excitation wavelength of 400 nm. Here, the excited photon energy is much larger than the band gap of Sb_2Se_3 thin film, therefore electrons can directly transfer from valence band to conduction band by absorbing an excited photon. Moreover, we propose a three-level approximation model to explain the two different absorption mechanism and further validate its rationality by numerical calculation. All the results indicate that Sb_2Se_3 is an ideal NOA material. Importantly, the combination of RSA and SA, which merely by thermal treatment regulation through one kind of material with identical chemical composition will find huge potential in future applications of ultrafast nonlinear optical devices such as optical diode, all-optical parallel logic gates, optical switches and optical power limiters.

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